

Improving the Performance of Brain-Computer Interface through Meditation Practicing

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Abstract—Cognitive tasks using motor imagery have been used for generating and controlling EEG activity in most brain-computer interface (BCI). Nevertheless, during the performance of a particular mental task, different factors such as concentration, attention, level of consciousness and the difficulty of the task, may be affecting the changes in the EEG activity. Accordingly, training the subject to consistently and reliably produce and control the changes in the EEG signals is a critical issue in developing a BCI system. In this work, we used meditation practice to enhance the mind controllability during the performance of a mental task in a BCI system. The mental states to be discriminated are the imaginative hand movement and the idle state. The experiments were conducted on two groups of subject, meditation group and control group. The time-frequency analysis of EEG signals for meditation practitioners showed an event-related desynchronization (ERD) of beta rhythm before imagination during resting state. In addition, a strong event-related synchronization (ERS) of beta rhythm was induced in frequency around 25 Hz during hand motor imagery. The results demonstrated that the meditation practice can improve the classification accuracy of EEG patterns. The average classification accuracy was 88.73% in the meditation group, while it was 70.28% in the control group. An accuracy as high as 98.0% was achieved in the meditation group.

I. INTRODUCTION

For several years, many efforts have been done to use the electro-encephalogram (EEG) as a new communication channel between human brain and computer. This new communication channel is called EEG-based brain-computer interface (BCI). Most of these efforts have been dedicated to the improvement of the accuracy and capacity of this EEG-based communication channel. In addition to employment of different signal processing approaches [1]-[3], some researchers have investigated the role of biofeedback [4], response verification [5], and mental practice [6] on EEG control.

The motor imagery is the essential part of most EEG-based communication systems. Motor imagery is defined as a dynamic state during which representation of a given motor act is internally rehearsed in working memory without any overt motor output. At this point one important question arises and that is how to imagine? Does the quality of mental rehearsal and the degree of imagined effort affect the performance of EEG-based BCI?

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During the performance of a cognitive task, there are many factors outside the motor imagery process that may be affecting the changes in EEG signals. There are now repeated evidences that mental state has significant effects on spatial-temporal EEG patterns. It has been known that the signal changes related to alertness, arousal, focused attention and sustained mental effort, cognitive load, and emotional state of the subject are present in EEG [7], [8]. Low brainwaves are associated with tuning off and lack of concentration, while fast brainwaves are associated with focused attention and sustained mental effort.

Mental practice will also affect the EEG signals produced during performance of motor imagery [6]. It is now well established that mental practice using motor imagery improves motor skills [9], but performance efficiency depends on the individual differences and imagery quality. Mental practice refers to repeating a physical skill in the mind, without any physical movement of the body, with the intent of learning or refinement. It was suggested that the mental rehearsal duplicates the actual motor pattern that is being rehearsed. The effects of mental practice on motor skill learning are the result of practice on central motor programming. According to this view, it seems logical that mental practice should modify the neuronal activity in the primary sensorimotor areas and consequently change the performance of EEG-based BCI.

It has been already demonstrated that the mental practice and concentration skills can improve the classification accuracy of the EEG patterns during imaginative hand movements. It was found that mental training has a significant effect on the classification accuracy over the primary motor cortex and frontal area [6].

In this work, we are going to evaluate the impact of meditation practice on EEG activity during hand movement imagination and examine the role of mental training on EEG control.

II. MEDITATION PRACTICE AND BCI

One important issue in controlling EEG activity to drive BCI using cognitive tasks is mind control. The subject should be able to consistently and reliably control the EEG activity. On the other hand, during the performance of a particular mental task (e.g., motor imagery), different factors such as concentration, attention, state of consciousness and the difficulty of task affect the changes in EEG activity. Due to the fact that internalized attention and mindfulness are two major core factors of mind behavior during the performance of a particular mental task, meditation practice

could enable the subject to consistently and reliably produce and control changes in the EEG activity.

Meditation is the natural process of withdrawing attention from externals, including from physical and mental processes, and consciously directing it inward to a chosen focus of concentration. Meditation is commonly considered to be the willful manipulation of one's state of mind and intentional self-regulation of attention. Meditation has been found in numerous studies to reduce experience of anxiety, and increase power of concentration.

In recent years, there have been various studies on characterizing the internalized attention and mindfulness as two major core factors of mind behaviors during meditation using psychophysiological parameters such as EEG [10] and autonomic nervous activity [11]. In practitioners of meditation, EEG recordings showed a distinct pattern including an increase in the power of alpha and theta band during meditative state of the mind [12]. Enhancement of alpha and theta activity has been historically tied to meditation and attributed to increased brain "idling". Moreover, it was suggested that meditative experience, characterized by less complex dynamics of the EEG, involves "switching off" irrelevant networks for the maintenance of focused internalized attention and inhibition of inappropriate information [13].

Meditation practice could enable the subject to increase his/her power of concentration, and to induce different states of consciousness and "thoughtless awareness" during the performance of a mental task.

III. EXPERIMENTAL PARADIGM

The current study utilized a two group experimental design, a meditation group, and a control group, each containing five subjects. The subjects with no experience of meditation served as the control group and did not receive any special training.

Meditation group consisted of five Transcendental Meditation (TM) practitioners with about one year experience. In transcendental meditation (TM), normally practiced for periods of twenty minutes twice a day, the subject is instructed to sit quietly with eyes closed and is then taught to repeat a certain type of sound or "mantra" according to a particular definite set of instructions. The mantras are a set of short speech sounds, meaningless in themselves, preserved by an ancient vedic tradition and assigned to individuals by the instructor on the basis of a set of objective rules which is trained to apply. They are chosen so as to "resonate" with the structure of an individual nervous system.

The EEG data were recorded at a sampling rate of 256 Hz by Ag/AgCl scalp electrodes placed according to the International 10-20 system. The eye blinks were recorded by placing an electrode on the forehead above the left brow line. The signals were referenced to the right earlobe. During

each experimental trial, one task was performed without any warning tone. Depending on the visual stimulus cue which was appeared on the monitor of computer at 2 s, the subject imagines either right-hand grasping or right-hand opening. If the visual stimulus is not appeared, the subject does not perform a specific task. Data were recorded for 5 s during each experimental trial and each trial was repeated 50 times for each task. There were three separate experiment days on each subject.

IV. RESULTS AND DISCUSSION

A. Time-Frequency Analysis

Event-related spectral perturbation (ERSP) analysis [14] was used for spectral analysis of the EEG signals. To compute an ERSP, baseline spectra are estimated from the EEG preceding the visual cue. The EEG obtained during each experimental trial is divided into short, overlapping data windows; and a moving average of the amplitude spectra of these is created. The obtained spectrum is then normalized by dividing by the mean baseline spectra. Normalized spectral for many trials are then averaged to obtain an average ERSP [14].

Examples of time-frequency distribution of EEG signals during hand movement imagination for two TM practitioners were shown in Fig. 1. These two subjects have practiced TM for more than one year. A strong broad-banded event-related synchronization (ERS) of beta rhythm in frequency around 25 Hz during hand motor imagery is observed. In addition, an event-related desynchronization (ERD) pattern was induced before imagination during resting state. Short-lasting beta bursts (beta rebound, beta ERS) were observed after imagination of hand, foot or tongue movement in able-bodied subjects with no meditation experience [15]. Further, pre-movement ERD and post-ERS were also found in a group of normal subjects during self-paced voluntary thumb, index finger and wrist movement [16]. In TM practitioner ND [Fig. 1 (a)], a post-imagination event-related synchronization (ERS) in alpha band with frequencies around 10 Hz was occurred at electrode positions F3, F4, Fz, C3, and T5. It is worthy to note that the hand mu rhythm blocked or desynchronized was reported at electrode positions C3 and C4 during hand motor imagery, whereas an enhancement of the hand area mu rhythm was observed during foot or tongue motor imagery [17].

The results indicate that there is a pre-imagination ERD in the alpha band focused around 10 Hz at positions Fz and C3. The pre-movement ERD of alpha and beta has been already reported during voluntary movement [18]. In addition, alpha ERS responses (6–12 Hz) have been observed in association with, e.g., attention, alertness and semantic memory and beta frequency responses have been associated with the activity of the motor cortices and recently also with also cognitive processing [19].

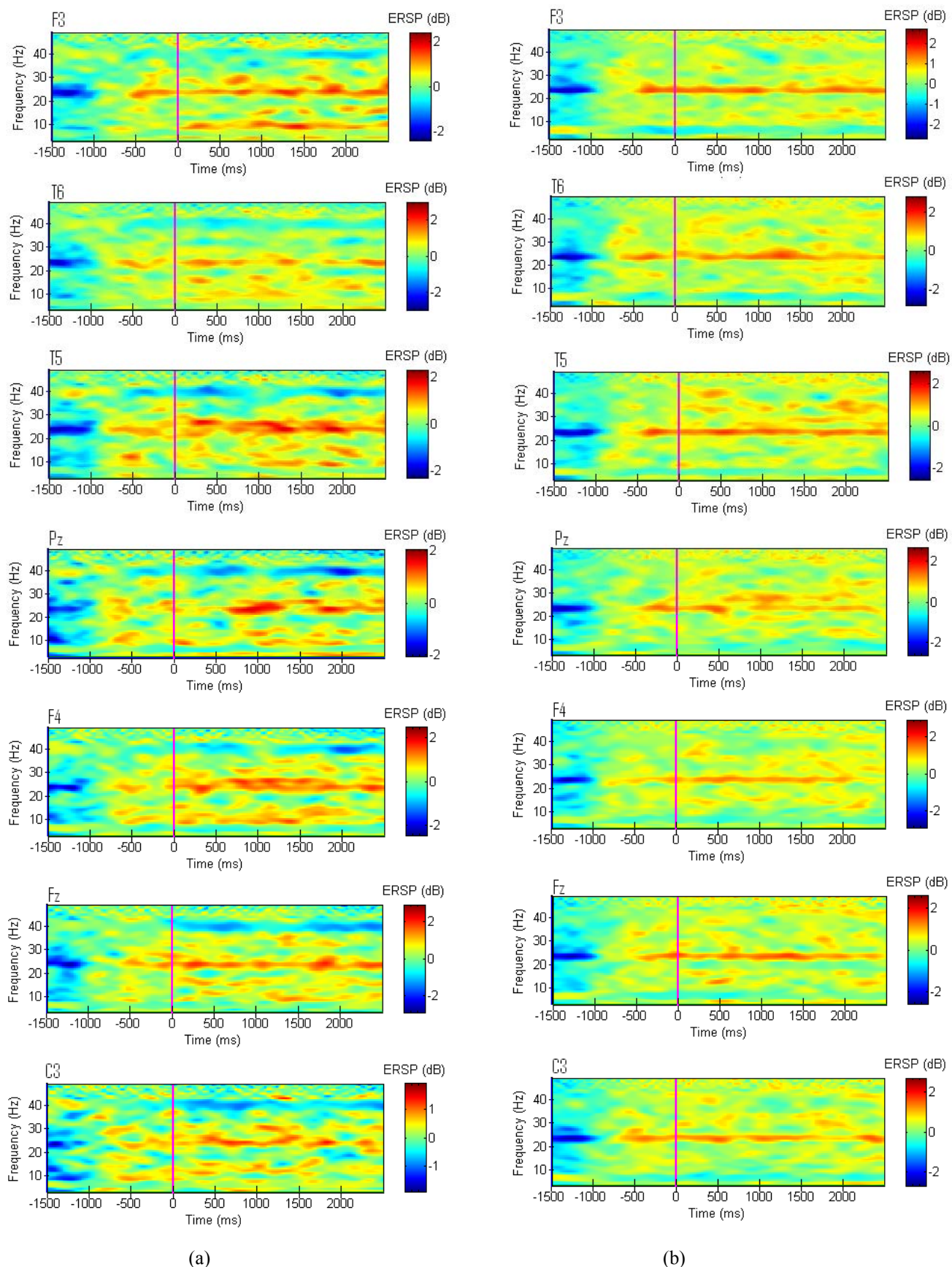


Fig. 1. Mean event-related spectral perturbation of the EEG signals during hand movement imagination in two TM practitioners ND (a) and MP (b).

B. Feature Extraction and EEG Classification

Original features are formed from 1-s interval of single-channel EEG data, in the time period 2.2-3.2 s, during each experimental trial. The window starting 0.2 s after cue presentation is used for classification because it takes time for the subjects to start imagination. The mean absolute value (MAV), variance, zero crossing and number of extrema of each interval, 5 AR parameters, energy of 8 wavelet subbands, 1-Hz frequency components between 1 and 45 Hz constitute the full set of features. The classifier is trained to distinguish between rest state and imaginative hand movement. The imaginative hand movement can be hand closing or hand opening.

From 200 data sets, 100 sets are randomly selected for network training, while the rest is kept aside for validation purposes.

In this work, for each EEG channel, a separate classifier was trained and tested. The desired output of each classifier is -1 or +1. The diagonal linear discrimination analysis (DLDA) [17] is here considered as the classifier. The classifier is trained to distinguish between rest state and imaginative hand movement. The output of classifiers is added and the *signum function* is used for computing the final decision.

Tables I and II summarize the results of classification accuracy of the EEG signals for all subjects in control and meditation group. It is observed that the best average classification accuracy obtained is 78.23% for control group, while it is 98.0% in meditation group. The average classification accuracy over all experiment days and subjects is 88.73% in meditation group which is 18.45% better than that obtained in control group.

The results demonstrated that the meditation practice can improve the performance of BCI. The analysis of variance (ANOVA) shows that the mean classification accuracies achieved in control and meditation group are significantly different ($p < 0.001$).

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TABLE I
CLASSIFICATION ACCURACY RATE OF EEG SIGNALS DURING HAND
MOVEMENT IMAGINATION IN CONTROL GROUP.

Subject	AJ	AL	ML	OV	SN	mean
Rate (%)	67.23	68.40	78.23	66.03	71.53	70.28

TABLE II
CLASSIFICATION ACCURACY RATE OF EEG SIGNALS DURING HAND
MOVEMENT IMAGINATION IN MEDITATION GROUP.

Subject	FM	MV	PE	MP	ND	mean
Rate (%)	87.53	96.29	98.00	86.50	75.35	88.73